



FYP PROJECT EXHIBITION & JOB FAIR 2026

INNOVATE • DETECT • PROTECT

CHASHM AI



REAL-TIME OBJECT DETECTION USING YOLOv8 & TFLITE

An efficient and lightweight object detection system
for real-time applications on Android devices.



GROUP MEMBERS

- Muhammad Okasha
- Muhammad Umar
- Sajjad Ahmad



SUPERVISED BY

Dr. Madiha Sher

KEY FEATURES



Real-Time
Detection



YOLOv8
Model



TFLite
Optimization



Android
Application



Lightweight
& Efficient



ESP32-CAM
Hardware

TECHNOLOGIES USED



YOLOv8



TensorFlow Lite



Android

PERFORMANCE RESULTS



mAP@0.5

94.6%



PRECISION

93.8%



RECALL

92.7%

ABOUT CHASHM AI

Chashm AI is a portable, real-time object detection solution that leverages the power of YOLOv8 and TensorFlow Lite to deliver high-accuracy results directly on Android devices. Built with lightweight hardware (ESP32-CAM) and optimized models, it ensures fast, efficient, and on-device inference for various real-world applications.



SCAN TO WATCH
DEMO VIDEO



SCAN TO VIEW
GITHUB REPOSITORY

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PROBLEM STATEMENT



Real-time object detection on resource-constrained Android devices is challenging due to limited processing power, high latency, and the need for lightweight yet accurate models.

OBJECTIVES



Develop a real-time object detection system using YOLOv8.



Optimize and deploy the model on Android using TensorFlow Lite.



Integrate ESP32-CAM for real-time video capture and inference.



Ensure the system is lightweight, efficient and suitable for real-world applications.

APPLICATIONS



Surveillance & Security



Industrial Automation



Traffic Monitoring



Robotics & Embedded AI



METHODOLOGY



Data Collection



Model Training (YOLOv8)



Model Optimization (TFLite)



Deployment on Android



Real-Time Inference

SYSTEM ARCHITECTURE

ESP32-CAM



- Captures Real-Time Video
- Streams Frames to Android

Wi-Fi

ANDROID DEVICE



- Runs YOLOv8 via TensorFlow Lite
- Performs Real-Time Object Detection
- Displays Results

OUTPUT



- Live Detection
- Bounding Boxes
- Class Labels
- Confidence Scores

TECH STACK

Y YOLOv8



TensorFlow Lite



Android



ESP32-CAM

HARDWARE OVERVIEW

ESP32-CAM



- Compact
- Low Power
- Built-in Camera
- Wi-Fi Enabled



Android Device

- Inference Engine
- TFLite Runtime
- Real-Time Display



Lightweight & Efficient Design

- Low Latency
- High Accuracy
- On-Device Inference

PERFORMANCE RESULTS



mAP@0.5

94.6%

High mean Average Precision indicates excellent overall detection performance.



PRECISION

93.8%

The system accurately identifies relevant objects with minimal false positives.



RECALL

92.7%

The system effectively detects most of the relevant objects in the frame.

LIVE DETECTION PREVIEW (INDOOR)



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GITHUB REPOSITORY

Scan to view the project repository and code.



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